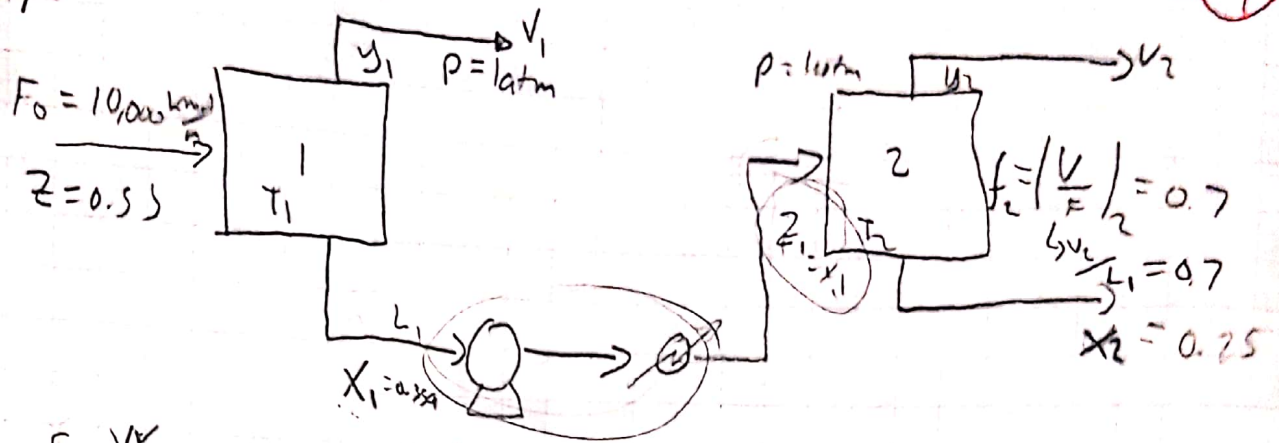


D2)



a)  $f = V/F_0$

Sep 1:  $F_0 = V_1 + L_1$   
 $z F_0 = y_1 V_1 + x_1 L_1$

$y_1 = -\left(\frac{1-f_1}{f_1}\right)x_1 + \frac{z}{f_1}$

$f_1 y_1 = (f_1 - 1)x_1 + z$

$f_1 y_1 = f_1 x_1 = -x_1 + z$

$f_1 (y_1 - x_1) = -x_1 + z$

$f_1 = \frac{-x_1 + z}{y_1 - x_1}$

$x_1 = 0.504$

graph  $y_1 = 0.78$

$f_1 = \frac{-0.504 + 0.55}{0.78 - 0.504}$

$f_1 = 0.151$  ✓

Sep 2:  $L_1 = V_2 + L_2$

$z_1 L_1 = y_2 V_2 + x_2 L_2$

$y_2 = \left(\frac{L_2}{V_2}\right)x_2 + z_1 \left(\frac{L_1}{V_2}\right)$

$y_2 = -\left(\frac{1-f_2}{f_2}\right)x_2 + \frac{z_1}{f_2}$

$y_2 = \left(\frac{1-0.7}{0.7}\right)x_2 + \frac{z_1}{0.7}$

$y_2 = 0.4286 x_2 + 1.4286 z_1$

$y_2 = -\frac{3}{7} x_2 + \frac{10}{7} z_1$

Graph  $x_2 = 0.25$   
 $y_2 = 0.62$  ✓

$z_2 = \frac{y_2 + 3/7 x_2}{10/7}$

$z_2 = \frac{(0.62) + 3/7 (0.25)}{10/7}$

$z_2 = 0.509 = x_1$  ✓

Interpolation

$x_1 = 0.509$   
 $T_1 = ?$

$x_0 = 50$

$T_0 = 73.1$

$x_2 = 60$

$T_2 = 71.2$

$T_1 = T_0 + (x_1 - x_0) \frac{T_2 - T_0}{x_2 - x_0} \Rightarrow T_1 = 72.929$  ✓

$x_2 = 0.25$

$T_2 = ?$

$x_4 = 0.2$

$T_4 = 81.7$

$x_3 = 0.3$

$T_3 = 78$

$T_2 = T_1 + (x_2 - x_1) \frac{T_3 - T_1}{x_3 - x_1} \Rightarrow T_2 = 79.85$  ✓

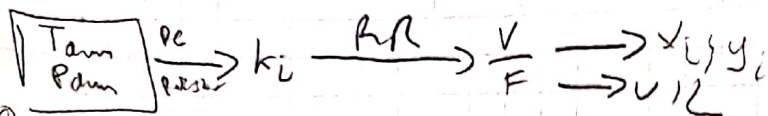
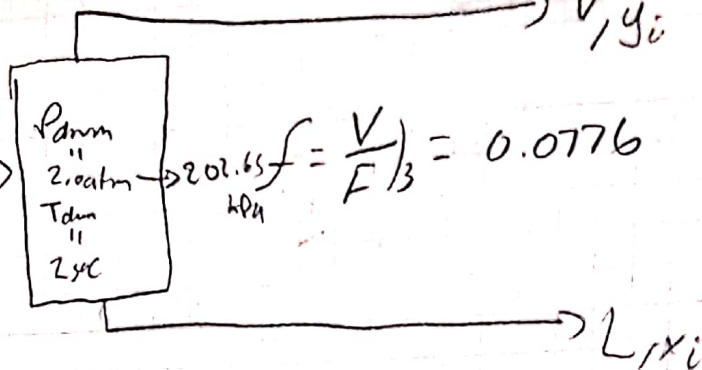
D8)

$$F_0 = 2000 \text{ kmol/h}$$

$$Z_1 = 0.05$$

$$Z_2 = 0.1$$

$$Z_3 = 0.85$$



①  $\left. \begin{matrix} k_1 = 74 \\ k_2 = 4.2 \\ k_3 = 0.12 \end{matrix} \right\} \text{De priester}$

② R-R eqn  $\rightarrow V/F = ?$

$$f(V/F) = \sum_{i=1}^3 \frac{(k_i - 1)Z_i}{1 + (k_i - 1)V/F} = \frac{(k_1 - 1)Z_1}{1 + (k_1 - 1)V/F} + \frac{(k_2 - 1)Z_2}{1 + (k_2 - 1)V/F} + \frac{(k_3 - 1)Z_3}{1 + (k_3 - 1)V/F}$$

$$(1) \quad f(V/F) = \frac{3.65}{1 + 73(V/F)} + \frac{0.32}{1 + 3.2(V/F)} + \frac{-0.748}{1 + 0.8(V/F)}$$

Newtonian Convergence

$$\frac{df}{d(V/F)} = - \sum_{i=1}^3 \frac{c (k_i - 1)^2 Z_i}{[1 + (k_i - 1)V/F]^2}$$

$$= - \left[ \frac{266.45}{(1 + 73(V/F))^2} + \frac{1.024}{(1 + 3.2(V/F))^2} + \frac{0.658}{(1 + 0.8(V/F))^2} \right] \quad (2)$$

Trial & Error  $V/F = 0.1$

$$\left. \frac{df}{d(V/F)} \right|_{0.1} = -5.246 \rightarrow f(0.1) = -0.137$$

$$\begin{aligned} \text{2nd } (V/F)_2 &= [V/F]_1 - \frac{f_1}{df/d(V/F)} \\ &= 0.1 - \frac{(-0.137)}{-5.246} = 0.0739 \end{aligned}$$

$$\rightarrow \left. \frac{df}{d(V/F)} \right|_2 = -7.938 \rightarrow f(0.0739) = 0.0296$$

$$(V/F)_3 = [V/F]_2 - \frac{f_2}{df/d(V/F)} = 0.0776$$

$$3d) f(0.0776) = 0.0017 < 0.005$$



$$\neq V/F$$

$$V = f F$$

$$= (2000)(0.0776)$$

$$V = 155.2 \text{ kmol/h}$$

$$F_0 = D + L$$

$$L = F_0 - V$$

$$L = 1844.8 \text{ kmol/h}$$

$$X_i = \frac{Z_i}{(1 + (k_i - 1)V/F)}$$

$$y_i = k_i X_i$$

$$X_1 = \frac{0.05}{(1 + (74 - 1)0.0776)} = 0.0075$$

$$y_1 = (74 \cdot 0.0075) = 0.555$$

$$X_2 = \frac{0.1}{(1 + (4.2 - 1)0.0776)} = 0.0801$$

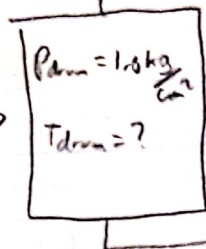
$$y_2 = (4.2 \cdot 0.0801) = 0.337$$

$$X_3 = \frac{0.85}{(1 + (0.12 - 1)0.0776)} = 0.912$$

$$y_3 = (0.12 \cdot 0.912) = 0.109$$

(Q9) Ethanol-water

$F = 1000 \text{ kg/h}$   
 $T_0 = 200^\circ\text{C}$   
 $\text{wt}\% = 0.3 \text{ Ethanol}$



$V, y_i$   
 $C_{p,V,i} = 14.66 + 3.758 \times 10^{-2} T = 2.091 \times 10^{-3}$   
 $C_{p,V,w} = 7.88 + 0.72 \times 10^{-2} T = 0.04533 \times 10^{-3} T$   
 $\left[ \frac{\text{kcal}}{\text{kmol}^\circ\text{C}} \right]$

$L, x_i$   
 $C_{p,L,\text{ethanol}} = 37.96 \text{ } T = 100^\circ\text{C}$   
 $C_{p,L,w} = 18.0 \frac{\text{kcal}}{\text{kmol}^\circ\text{C}}$

E.B.  $Fh_F + Q_{\text{loss}} = Vh_V + Lh_L$

30wt% Assumed 100 kg  $\text{wt}\% \rightarrow Z$   
 $\text{Ethanol } 30\text{wt}\% \rightarrow 30 \text{ kg Ethanol} \times \frac{1 \text{ kmol}}{46.07 \text{ kg}} = 0.6512 \text{ kmol Ethanol}$   
 $\text{Water } 70\text{wt}\% \rightarrow 70 \text{ kg Water} \times \frac{1 \text{ kmol}}{18.016 \text{ kg}} = 3.885 \text{ kmol Water}$   
 $\text{Total moles} = 4.5366 \text{ kmol}$   
 $Z = 0.1435$   
 $Z_{\text{water}} = 0.8565$   
 $M_{\text{water}} = 6.512 \text{ kmol} \times 0.1435 \times 18.016 = 1.68 \text{ kmol}$   
 $M_{\text{ethanol}} = 0.6512 \text{ kmol} \times 0.8565 \times 46.07 = 2.56 \text{ kmol}$   
 $M_{\text{total}} = 4.24 \text{ kmol}$

$\Rightarrow h_F(T_F, Z) = Z_A C_{p,A} (T_F - T_{\text{ref}}) + Z_B C_{p,B} (T_F - T_{\text{ref}})$   
 $= (0.1435)(37.96)(200^\circ\text{C} - 0) + (0.8565)(18.0)(200^\circ\text{C} - 0)$   
 $h_F = 4,172.85 \frac{\text{kcal}}{\text{kmol}} \times \frac{1 \text{ kmol}}{(0.1435 \times 46.07 + 0.8565 \times 18)} = 189 \frac{\text{kcal}}{\text{kg}}$

$h_F = \frac{\text{kcal}}{\text{kg}} \Rightarrow \text{Liquid}$   
 $\text{wt}\% = 0.3$   
 $\text{Vapor}$   
 $\text{wt}\% = 0.15 \text{ Ethanol}$   
 $\text{wt}\% = 0.62 \text{ Ethanol}$   
 $T_{\text{drum}} = 88^\circ\text{C}$

$F = V + L$

$1000 = L = V$

$V = 319.1489 \text{ kg/h}$

$F_{\text{wt Ethanol}} = V_{\text{wt V}} + L_{\text{wt L}}$

$1000(0.3) = V(0.62) + L(0.15)$

$300 = (1000 - L)(0.62) + 0.15L$

$300 = 620 - 0.62L + 0.15L$

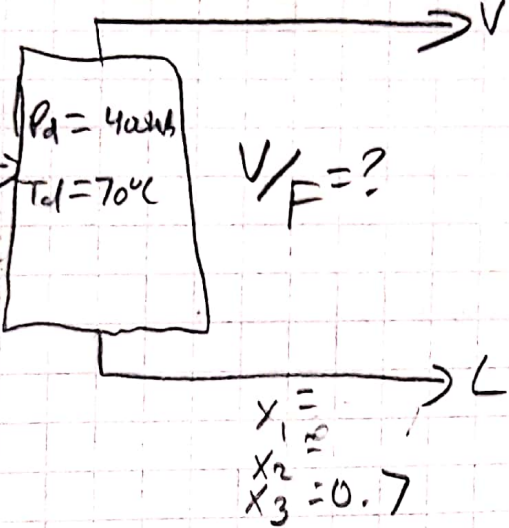
$L = 680.88 \text{ kg/h}$



David

39)

$z_1 = 0.35$  butane  
 $z_2 = ?$  propane  
 $z_3 = ?$  n-hex



$k_1 = 5$   
 $k_2 = 1.9$   
 $k_3 = 0.3$

Pe-Fresh ch-t

RR  $\rightarrow f(V/F) = \sum_{i=1}^3 \frac{(k_i - 1)z_i}{1 + (k_i - 1)V/F} = 0$

$V_D = \frac{z_D}{(1 + (k_D - 1)V/F)}$

$\sum z_i = 1$

$\sum x_i = 1$

Solver RR:  $\sum_{i=1}^3 \frac{(k_i - 1)z_i}{1 + (k_i - 1)V/F} = 0$

Note  $\sum z_i = 1$   
 $z_2 = 1 - z_1 - z_3$

$X_{cb} = \frac{z_3}{(1 + (k_3 - 1)V/F)} = 0.7$

By changing  $z_3, V/F$

$\Rightarrow \{ z_3 = z_{cb} = 0.22$

$\{ V/F = 0.979592$

X

— )

AY McCabe Thiele diagram

$V, y_i$ ?

$F = 100 \text{ kmol/h}$

$T_d = 50^\circ\text{C}$   
 $P_d = 200 \text{ kPa}$

$z_1 = 0.20$   
 $z_2 = 0.10$   
 $z_3 = 0.15$   
 $z_4 = 0.45$

$L, x_i$ ?

$\alpha_{T_d, P_d} \rightarrow$   
 $k_1 = 7.0$   
 $k_2 = 2.40$   
 $k_3 = 0.80$   
 $k_4 = 0.30$

② RR  $\rightarrow V/F$

$$f(V/F) = \sum \frac{(k_i - 1)z_i}{1 + (k_i - 1)V/F} = 0$$

Goal seek w RR = 0

Excel  $\Rightarrow V/F = 0.5113$

③  $V, L$  from  $V/F$

$$V/F = C$$

$$V = CF$$

$$F = V + L$$

$$L = F - V$$

Excel  $\Rightarrow$   
 $V = 511.2998$   
 $L = 488.7002$

$$④ x_i = \sum \frac{z_i}{1 + (k_i - 1)V/F}$$

$$y_i = k_i z_i$$

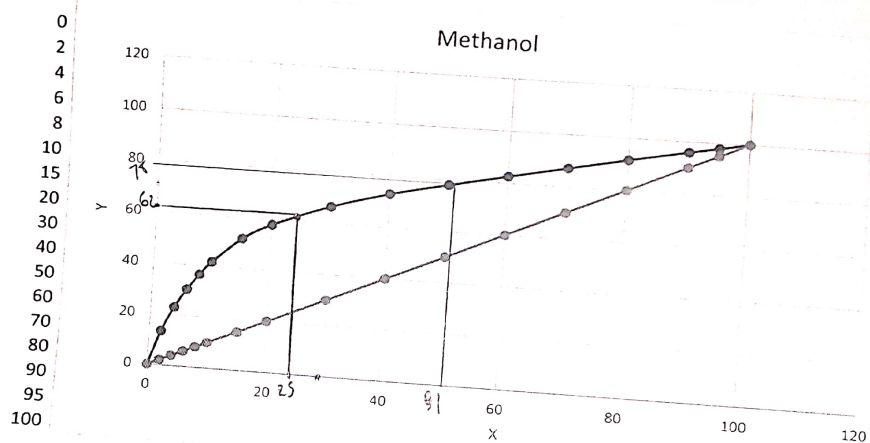
Excel  $\Rightarrow$   
 $x_1 = 0.07375$   
 $x_2 = 0.058781$   
 $x_3 = 0.167$   
 $x_4 = 0.701$

$y_1 = 0.516$   
 $y_2 = 0.1398$   
 $y_3 = 0.1337$   
 $y_4 = 0.210$

HW # 1 - David Alvarado  
A1)

9

| Methanol liquid | Methanol Vapor | y |
|-----------------|----------------|---|
| 0               | 0              |   |
| 2               | 13.4           |   |
| 4               | 23             |   |
| 6               | 30.4           |   |
| 8               | 36.5           |   |
| 10              | 41.8           |   |
| 15              | 51.7           |   |
| 20              | 57.9           |   |
| 30              | 66.5           |   |
| 40              | 72.9           |   |
| 50              | 77.9           |   |
| 60              | 82.5           |   |
| 70              | 87             |   |
| 80              | 91.5           |   |
| 90              | 95.8           |   |
| 95              | 97.9           |   |
| 100             | 100            |   |



HW # 4 - David Alvarado  
A2) Matlab

(10)

1-7

\$

k1 5  
k2 1.9  
k3 0.3

z1 0.35  
z2 0.429332  
z3 0.22

V/F 0.979592

x1 0.071162  
x2 0.22817  
x3 0.7

RR1

RR1

RR2

RR3

RR

Eq

Equal to

0

0.284647

0.205353

-0.49

-1.6E-07

0

x

0.7

0.7



|     |          | Equation | Equal to |
|-----|----------|----------|----------|
| k1  | 7        | RR1      | 0.4425   |
| k2  | 2.4      | RR2      | 0.081594 |
| k3  | 0.8      | RR3      | -0.03342 |
| k4  | 0.3      | RR4      | -0.49059 |
|     |          | RR       | 9.08E-05 |
| z1  | 0.3      |          |          |
| z2  | 0.1      |          |          |
| z3  | 0.15     |          |          |
| z4  | 0.45     |          |          |
| V/F | 0.5113   |          |          |
| F   | 1000     |          |          |
| L   | 488.7002 |          |          |
| V   | 511.2998 |          |          |
|     |          | x1       | 0.07375  |
|     |          | x2       | 0.058281 |
|     |          | x3       | 0.167086 |
|     |          | x4       | 0.700836 |
|     |          | y1       | 0.51625  |
|     |          | y2       | 0.139875 |
|     |          | y3       | 0.133669 |
|     |          | y4       | 0.210251 |